

# Supplementary Material S1

## Dirac-Expanded Bracket Notation for the Controlled- $X$ Gate

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Supplement to “Exact Channel Equivalence and Leakage-Aware Same-Backend Tomography for a Five-Qubit Coherent EPR-State Transfer Network with a GHZ-Type Resource Family”

### Purpose and relation to the main manuscript

This supplementary note contains the extended controlled- $X$  derivation that was formerly included as Appendix B of the main manuscript. The title and notation in this version are synchronized with the current main manuscript; the hardware tomography and CP-TNI reconstruction are documented separately in Supplementary File S2. It is separated from the article because the result is supporting notation rather than a central contribution of the channel-equivalence theorem. The derivation fixes the computational-basis convention, gives a compact Dirac-expanded representation of  $CX_{i,j,q}$ , embeds the operator explicitly in an arbitrary  $q$ -qubit tensor product, and checks the two- and three-qubit conventions used by the manuscript. The notation is motivated by the Dirac extension of CNOT gates discussed by Yamanashi, Ozawa, and Okazaki [1].

### 1 Dirac-expanded controlled- $X$ representation

The controlled- $X$  gate acting on a  $q$ -qubit register is denoted by

$$\boxed{CX_{i,j,q}}, \quad (S1)$$

where  $i$  denotes the control qubit,  $j$  the target qubit, and  $q$  the total number of qubits, with

$$0 \leq i, j \leq q-1, \quad i \neq j. \quad (S2)$$

The computational-basis ordering used throughout the main manuscript is

$$|q_{q-1} \cdots q_1 q_0\rangle = |q_{q-1}\rangle \otimes \cdots \otimes |q_1\rangle \otimes |q_0\rangle, \quad q_k \in \{0, 1\}. \quad (S3)$$

Circuit wires are displayed from top to bottom in ascending index order  $q_0, q_1, \dots, q_{q-1}$ . Hence the rightmost tensor factor  $|q_0\rangle$  corresponds to the top wire and the leftmost tensor factor  $|q_{q-1}\rangle$  to the bottom wire.

For a binary variable  $x_k \in \{0, 1\}$ , let  $\bar{x}_k = 1 - x_k$ . The binary power of the Pauli- $X$  operator on qubit  $k$  is

$$\boxed{X_k^{x_k} = |0\rangle_k \langle x_k|_k + |1\rangle_k \langle \bar{x}_k|_k}. \quad (S4)$$

Thus

$$X_k^0 = I_k = |0\rangle \langle 0| + |1\rangle \langle 1|, \quad (S5)$$

$$X_k^1 = X_k = |0\rangle \langle 1| + |1\rangle \langle 0|, \quad (S6)$$

and, for  $x_k, y_k \in \{0, 1\}$ ,

$$\boxed{X_k^{x_k} |y_k\rangle = |y_k \oplus x_k\rangle} = \begin{cases} |y_k\rangle, & x_k = 0, \\ |\bar{y}_k\rangle, & x_k = 1. \end{cases} \quad (\text{S7})$$

For  $\text{CX}_{i,j,q}$ , the binary value of the control qubit determines whether the target is acted on by  $I_j$  or  $X_j$ . Restricting first to the control and target factors gives

$$\boxed{\sum_{x_i=0}^1 |x_i\rangle_i \langle x_i|_i X_j^{x_i}} = |0\rangle_i \langle 0|_i I_j + |1\rangle_i \langle 1|_i X_j. \quad (\text{S8})$$

Identity action on all qubits  $k \neq i, j$  is understood at this stage.

Applying Equation (S4) to the target qubit yields

$$X_j^{x_i} = |0\rangle_j \langle x_i|_j + |1\rangle_j \langle \bar{x}_i|_j. \quad (\text{S9})$$

Substitution into Equation (S8) gives the compact Dirac-expanded bracket representation

$$\boxed{\text{CX}_{i,j,q} = \sum_{x_i=0}^1 |x_i\rangle_i \langle x_i|_i \left( |0\rangle_j \langle x_i|_j + |1\rangle_j \langle \bar{x}_i|_j \right)}, \quad (\text{S10})$$

again with identity action on all remaining qubits.

Expanding the sum gives

$$\begin{aligned} \text{CX}_{i,j,q} = & |0\rangle_i |0\rangle_j \langle 0|_i \langle 0|_j + |0\rangle_i |1\rangle_j \langle 0|_i \langle 1|_j \\ & + |1\rangle_i |0\rangle_j \langle 1|_i \langle 1|_j + |1\rangle_i |1\rangle_j \langle 1|_i \langle 0|_j. \end{aligned} \quad (\text{S11})$$

The first two terms correspond to control value 0 and leave the target unchanged, whereas the last two correspond to control value 1 and exchange the target states  $|0\rangle_j$  and  $|1\rangle_j$ .

## 2 Explicit embedding in a $q$ -qubit tensor product

For each  $x_i \in \{0, 1\}$ , define

$$A_k^{(x_i)} = \begin{cases} |x_i\rangle_i \langle x_i|_i, & k = i, \\ X_j^{x_i}, & k = j, \\ I_k, & k \neq i, j. \end{cases} \quad (\text{S12})$$

Then, consistently with Equation (S3), the fully embedded operator is

$$\boxed{\text{CX}_{i,j,q} = \sum_{x_i=0}^1 A_{q-1}^{(x_i)} \otimes A_{q-2}^{(x_i)} \otimes \cdots \otimes A_1^{(x_i)} \otimes A_0^{(x_i)}}. \quad (\text{S13})$$

Equation (S13) fixes the tensor-factor positions explicitly and remains valid for adjacent or nonadjacent qubits and for either  $i < j$  or  $i > j$ .

For a computational-basis input  $|q_{q-1} \cdots q_1 q_0\rangle$ , define

$$q'_k = \begin{cases} q_k, & k \neq j, \\ q_j \oplus q_i, & k = j. \end{cases} \quad (\text{S14})$$

Then

$$\boxed{\text{CX}_{i,j,q} |q_{q-1} \cdots q_1 q_0\rangle = |q'_{q-1} \cdots q'_1 q'_0\rangle}, \quad (\text{S15})$$

with

$$\boxed{q'_i = q_i, \quad q'_j = q_j \oplus q_i}, \quad (\text{S16})$$

and  $q'_k = q_k$  for  $k \neq i, j$ . Equivalently,

$$\boxed{X_j^{q_i} |q_j\rangle = |q_j \oplus q_i\rangle}. \quad (\text{S17})$$

### 3 Consistency checks used in the manuscript

For  $\text{CX}_{0,1,2}$ , the tensor ordering is

$$|q_1 q_0\rangle = |q_1\rangle \otimes |q_0\rangle, \quad (\text{S18})$$

and qubit 0 is the control while qubit 1 is the target. Hence

$$\boxed{\text{CX}_{0,1,2} |q_1 q_0\rangle = |(q_1 \oplus q_0) q_0\rangle}. \quad (\text{S19})$$

For the two-qubit input used in the main manuscript,

$$\text{CX}_{0,1,2} (|y\rangle_1 \otimes [\alpha |0\rangle_0 + (-1)^x \beta |1\rangle_0]) = \alpha |y, 0\rangle + (-1)^x \beta |\bar{y}, 1\rangle. \quad (\text{S20})$$

Thus the controlled- $X$  convention reproduces the encoded state definition used in the manuscript.

As a second check, consider  $\text{CX}_{0,2,3}$ . With

$$|q_2 q_1 q_0\rangle = |q_2\rangle \otimes |q_1\rangle \otimes |q_0\rangle, \quad (\text{S21})$$

qubit 0 is the control, qubit 2 the target, and qubit 1 is unchanged. Therefore

$$\boxed{\text{CX}_{0,2,3} |q_2 q_1 q_0\rangle = |(q_2 \oplus q_0) q_1 q_0\rangle}. \quad (\text{S22})$$

Consequently, the control-target expression

$$\boxed{\text{CX}_{i,j,q} = \sum_{x_i=0}^1 |x_i\rangle_i \langle x_i|_i X_j^{x_i}} \quad (\text{S23})$$

together with the explicit tensor embedding in Equation (S13) provides a single Dirac-expanded representation valid for arbitrary register size and qubit placement.

## References

- [1] Yamanashi, S.; Ozawa, K.; Okazaki, H. On Dirac extension bra-ket notations of CNOT gates. *IEICE Tech. Rep.* **2023**, *122*, CAS2022-61, 1–6. (In Japanese).